Yiwei Shu

Ph.D. Student in Robotics | Field and Planetary Robotics

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Education

Ph.D. — **University of Chinese Academy of Sciences**, Robotics and Space Robotics — Beijing, China, 2024 – present
- Affiliated with the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences.
- Research on autonomous exploration, legged field robots, active SLAM, motion planning, and planetary robotics.

B.Eng. — **ShanghaiTech University**, Electronic Information Engineering — Shanghai, China, Sept 2020 – June 2024
- Undergraduate research in robotics, visual control, point-cloud registration, medical imaging, and integrated-circuit design.

Exchange — **University of Illinois Urbana-Champaign**, Exchange Program — Urbana, IL, USA, Jan 2023 – May 2023
- Completed Quantum Physics, Data Science Discovery, and Statistical Analysis with A+ grades.

Manuscripts Under Review and Submitted

Finite-State Visual Servo Control for Autonomous Target Landing of Low-Cost Underactuated Fixed-Wing Aircraft — June 2026

Yiwei Shu, Shuquan Wang. Manuscript under review at *Aerospace Science and Technology*. Proposed an IMU/monocular-camera finite-state visual servo controller and validated it with CFD, randomized simulations, and real flight tests; reduced final image-plane radial error from 0.5049 to 0.0027 against a sliding-mode baseline.

From End-Effector Targets to Whole-Arm Postures: Configuration-Space Waypoint Recovery for Reactive Manipulation — July 2026

Yiwei Shu et al. Manuscript submitted to *IEEE Robotics and Automation Letters* (submission status: received). Introduced PostureWaypoint for online configuration-space recovery; evaluated four manipulators across six constrained 3D tasks and validated the method on an RM75 robot.

RKILPlanner: Risk-based Kinematic Imitation Learning Planning Towards Human-Like Driving in Highway Scenarios — 2026

Yiwei Shu, Shuquan Wang. Manuscript submitted to *ROBIO 2026*. Combined Driver's Risk Field maps, driver-style encoding, and LSTM/CNN features in a differentiable kinematic imitation-learning planner evaluated on the HighD naturalistic highway dataset.

Publications and Presentations

Expanding Grasp Reachability: Field-based Pose-constrained Two-stage Grasp Planning — July 2026

Yiwei Shu, Shuquan Wang, Jun Li, Hanying Sang, and Yidong Ye. Accepted full paper and oral presentation at *ICoSR 2026*. Developed reachability and manipulability fields with a two-stage policy that expands pose-constrained grasping near workspace boundaries; validated in simulation and on a 7-DoF RM75 manipulator.

Learning-based Extended-workspace Manipulation: A Self-relocating Space Crawler for Autonomous Deep-space Outposts — 2026

Yiwei Shu, Shuquan Wang. Poster presentation at the *46th COSPAR Scientific Assembly 2026*. Proposed a passive-rail, self-relocating space crawler and hierarchical reinforcement-learning framework for autonomous extended-workspace manipulation in deep-space outposts.

Attitude Control of Unpowered Aircraft in Flight Based on Kalman Filter Visual Servo-PID Optimization Algorithm — July 2025

Yiwei Shu, Shuquan Wang. Accepted for presentation and publication in the proceedings of the *44th Chinese Control Conference (CCC 2025)*. Combined visual servoing, adaptive PID control, and Kalman filtering for robust low-cost unpowered-aircraft attitude control.

Projects

Reliable Autonomous Exploration for Unitree Go2 2026 – present

Reliable active SLAM and navigation for quadruped robots in lunar-analogue and unstructured environments.

- Integrate LiDAR, depth sensing, grid-based mapping, and exploration planning for the Unitree Go2 platform.
- Build Isaac Lab lunar-terrain benchmarks and real-physics evaluation pipelines for legged navigation.
- Investigate calibrated world models, distribution shift, terrain traversability, and safe sim-to-real deployment.

PostureWaypoint for Reactive Whole-Arm Manipulation 2025 – present

Configuration-space waypoint recovery for online manipulation in cluttered and constrained workspaces.

- Resolve end-effector waypoint ambiguity through whole-arm clearance, orientation consistency, and joint-feasibility constraints.
- Evaluate on RM75, Franka Panda, UR5e, and KUKA iiwa manipulators with simulation and real-robot experiments.

Field-based Pose-constrained Two-stage Grasp Planning 2025 – 2026

Reachability- and manipulability-aware grasp planning for objects near manipulator workspace boundaries.

- Construct reachability and manipulability fields directly from the robot model.
- Use a sweep-then-pick policy when direct grasping is infeasible under strict pose constraints.

Autonomous Fixed-Wing Target Landing 2023 – 2026

Low-cost visual-servo control for underactuated and unpowered fixed-wing aircraft.

- Combine CFD-based aerodynamic characterization, IMU guidance, monocular visual servoing, and adaptive control.
- Validate controllers through randomized simulation and physical flight experiments.

Technical Skills

Programming: Python, MATLAB, C, C++, R, Verilog, LaTeX

Robotics: ROS/ROS 2, PCL, Unitree Go2, RM75, STM32, LiDAR and depth sensing

Simulation and Learning: Isaac Lab, MuJoCo, PyTorch, scikit-learn, numerical simulation

Engineering Tools: Git, Linux, SolidWorks, Altium Designer, ANSYS Fluent, Cadence, Vivado

Selected Honors

- Second Prize, East China Division, 6th National Undergraduate IC Innovation and Entrepreneurship Competition (2022)
- Second Prize, RoboMaster Eastern Division (2022)
- Third Prize, RoboMaster National Competition (2022)

Teaching and Leadership

- Teaching Assistant, Digital Circuits, ShanghaiTech University (September-December 2022): led tutorials and assisted with grading.
- Sentry Team Lead, RoboMaster (2021-2023): coordinated embedded programming, PCB design, mechanical modeling, and team planning.